X-ray techniques to study the structure of chemical compounds. In 1929, she worked out that the carbon atoms in benzene were arranged in a regular hexagonal ring. She went on to apply X-ray crystallography to the study of bladder stones and drugs used to relax muscles. In 1945, she became one of two women admitted as fellows of the Royal Society, London, which had previously only admitted men.

JOHN VON NEUMANN

1903–1957

Brilliant Hungarian-born mathematician John von Neumann worked on the design and development of early high-speed electronic digital computers. From 1930, he was based at Princeton, New Jersey, where—with American meteorologist Jule Charney—he put together the first computer-based weather predictions. In 1945, he worked out a model for a computer that could store programs, as well as data. All computers since then have been based on this model, which is known as von Neumann architecture.

WENDELL STANLEY

1904–1971

Born in Indiana, Wendell Stanley determined the molecular structure of viruses by using X-ray diffraction. He went on to share a Nobel Prize in Chemistry in 1946 for his research on the tobacco mosaic virus. During World War II, his insight into viruses as the cause of infectious disease aided the development of a vaccine against influenza. Shortly after the war, he moved to the University of California, Berkeley, and in 1954, he and his collaborators crystallized the polio virus—a major step in enabling researchers to develop a vaccine against the disease.

RITA LEVI-MONTALCINI

1909–2012

After struggles with her father, who believed women should stay at home, Rita Levi-Montalcini studied medicine at the University of Turin and went on to make important discoveries about nerve growth factors (NGFs). She moved to the US in 1947, and was made a professor at Washington University. Initially largely ignored, NGFs were later understood to be important in several diseases, including Alzheimer's, infertility, and cancer. Along with biochemist Stanley Cohen, who helped isolate an NGF, Levi-Montalcini was awarded the 1986 Nobel Prize in Physiology or Medicine.

RUBY PAYNE-SCOTT

1912–1981

Australian Ruby Payne-Scott had a short but productive career as a radio physicist in the 1940s and '50s. In June 1941, she was one of only two women physicists to be employed by the Radiophysics Laboratory at the University of Sydney. She helped develop radars for Australia's coastline in World War II, and from 1946 to 1951 was part of the team that developed a means of measuring radio emissions from the Sun and stars.

JONAS SALK

1914–1995

American virologist Jonas Salk developed the first safe and effective vaccine for polio. His vaccine used the killed virus—no live virus was injected. Early trials showed this vaccine gave immunity against the disease, but it was not long-lasting. In 1954, large-scale trials began. More than 1 million children were injected with the vaccine, which proved up to 90 percent effective against paralytic polio. At the same time, Albert Sabin was developing an oral vaccine against polio using live, attenuated (weakened) viruses. After another major trial, this vaccine was deemed successful, too. Easier to administer than the Salk vaccine, Sabin's oral vaccine had largely replaced Salk's by the early 1960s.

LYMAN SPITZER JR.

1914–1997

Renowned US astrophysicist Lyman Spitzer Jr. was the first scientist to conceive the idea of space telescopes. He had highlighted the problem of detecting nonvisible radiation through Earth's atmosphere. The solution that he proposed, in 1946, was to put a telescope into space. He understood the obstacles to such a proposal, which included the challenges of space travel and of designing an instrument capable of operating in space by remote control from Earth. In 1977, his campaign for a space telescope paid off, and Congress granted federal funding for the Hubble Space Telescope.

ROBERT EDWARDS

1925–2013

British physiologist Robert Edwards developed the techniques that allowed the first successful "test-tube" pregnancy. Edwards began working on fertilization in 1955 and spent 20 years creating the right conditions for an egg to be fertilized artificially, outside the womb. He began collaborating with surgeon Patrick Steptoe in 1968. Steptoe harvested eggs from female volunteers using keyhole surgery. The first IVF baby was born in July 1978, but Edwards had to wait over 30 years to be awarded a Nobel Prize in Physiology or Medicine in 2010.